

M09 Module Review

ETL/ELT + Data Quality + Transformation Engineering | Review after attempt

Review rule

Open only after a genuine attempt. A review is diagnostic material, not the first way to solve the task.

M09-L01 - ETL vs ELT and transformation boundaries

Strong evidence: You can justify transformation placement and state what evidence remains invariant across both patterns.

Repair focus: Selecting ETL/ELT from fashion instead of defining where raw evidence, quality checks and recovery live.

M09-L02 - Raw, staging, clean/intermediate and serving contracts

Strong evidence: You can place a new transformation in a layer and defend the placement.

Repair focus: Mixing destructive cleaning into raw or burying all business logic in one serving query.

M09-L03 - Deterministic transformations and production SQL style

Strong evidence: Repeated runs on unchanged input return identical rows/values and the grain is explicit.

Repair focus: Using max(updated_at) without a deterministic tie-breaker when source rows can tie.

M09-L04 - Deduplication and key integrity

Strong evidence: Every duplicate outcome follows a deterministic rule and row accounting reconciles.

Repair focus: Calling drop_duplicates() without documenting which record survives and why.

M09-L05 - Schema checks and data contracts

Strong evidence: Breaking drift is detected before target mutation and failure evidence is preserved.

Repair focus: Renaming sales_region to region automatically without an approved mapping and history.

M09-L06 - Source-to-target reconciliation

Strong evidence: Counts, keys and selected measures reconcile independently.

Repair focus: Declaring success because the serving count looks reasonable without checking key/value equivalence.

M09-L07 - Incremental processing and safe reruns

Strong evidence: Final serving state is stable across normal rerun/replay and all decisions are counted.

Repair focus: Using ingestion timestamp as winner when business update timestamp defines the source version.

M09-L08 - Failure behavior, retries, quarantine and atomic state

Strong evidence: Failure evidence is visible and recovery does not duplicate or skip business records.

Repair focus: Writing processed-file state before target/reconciliation finishes.

M09-L09 - Lineage, run metadata and auditability

Strong evidence: A reviewer can reconstruct the pipeline lineage from saved evidence.

Repair focus: Keeping only final CSV/table with no record of which source or transformation version produced it.

M09-L10 - Build and defend a reliable batch pipeline

Strong evidence: You can operate and explain the full scenario without the reference implementation open.

Repair focus: Finishing the happy path and skipping failure/replay validation.

M09-L11 - dbt-style project anatomy, sources and dependency graph

Strong evidence: A reviewer can follow dependencies without reading every SQL file.

Repair focus: Hard-coding downstream relation names so lineage/build order becomes hidden.

M09-L12 - Executable data tests, quality severity and CI boundaries

Strong evidence: Critical contract violations stop the build and produce actionable evidence.

Repair focus: Changing a failing test to warning because the data is inconvenient.

M09-L13 - Incremental models, materialization, documentation and handoff

Strong evidence: Another engineer can rebuild, test and safely rerun the transformation project.

Repair focus: An incremental model that works on first run but duplicates/loses rows on update/full refresh.